

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #47 - Developer Tools & IT Operations

Domain & SSL Certificate Expiry Alert System

1. Problem Context & Overview

An IT operations utility that performs automated daily WHOIS registration and SSL/TLS handshake audits, alerting SysAdmins through escalation ladders before domain or certificate expiration.

Target Stakeholders / Actors: SysAdmin, Security Officer

2. Functional Requirements (FR-001 - FR-005)

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall initiate TLS handshakes against every monitored domain/endpoint on a daily schedule, extract the SSL/TLS certificate's expiration (NotAfter) date, and raise escalating alerts at 30, 15, and 3 days before expiry.	High	Pass: An alert is queued for a certificate expiring in 15 days. Fail: An expired certificate is not flagged with any notification.	Certificate expiry is the single most common cause of unplanned outages and browser trust warnings; automated, time-boxed detection is the core value of the system.
FR-002	The system shall perform a daily WHOIS lookup for every monitored domain, extract the domain registration expiry date, and raise escalating alerts at 30, 15, and 3 days before the registration lapses.	High	Pass: An alert is generated when a domain's registration is due to lapse within 30 days. Fail: A domain nearing registration expiry produces no alert.	Unlike certificate expiry, a lapsed domain registration can cause total, hard-to-reverse loss of service (including email and DNS), so it must be tracked independently of SSL status.
FR-003	The system shall deliver alerts through configurable channels (e-mail, SMS, and chat-ops webhook) following an escalation ladder that widens the recipient list and urgency as the expiry threshold tightens (e.g., SysAdmin only at 30 days; SysAdmin + Security Officer at 15 and 3 days).	High	Pass: A 3-day alert reaches both SysAdmin and Security Officer on all configured channels within 60 seconds of being triggered. Fail: An escalation stage is skipped or the recipient list fails to expand as the deadline approaches.	A single notification channel or recipient is a single point of failure; a graduated ladder ensures the right people are pulled in as risk increases.
FR-004	The system shall let a SysAdmin or Security Officer acknowledge an active alert, recording the acknowledging user's ID and a timestamp, and shall suppress further re-notification for that specific alert instance once acknowledged.	Medium	Pass: An acknowledged alert is logged with actor and timestamp and does not re-fire on the next scan cycle. Fail: An acknowledged alert re-triggers, or the acknowledgment is not recorded in the audit log.	Provides accountability for who is handling an issue and prevents alert fatigue from repeated notifications for a problem already in progress.

ID	Description	Priority	Acceptance Criteria	Rationale
FR-005	The system shall generate a daily/weekly compliance report and an on-demand dashboard summarizing the status of all monitored domains and certificates (Healthy, Expiring Soon, Expired), accessible to SysAdmin and Security Officer.	Medium	Pass: The dashboard/report correctly categorizes every monitored asset by current risk status. Fail: The report omits an asset or misclassifies its expiry risk.	Gives stakeholders portfolio-wide visibility for audits and capacity planning, beyond the scope of any single alert.

3. Non-Functional Requirements (NFR-001 & NFR-002)

ID	Description	Type	Acceptance Criteria	Rationale
NFR-001	The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes, using only TLS 1.2+ handshakes with certificate-chain validation.	Performance & Security	Pass: Benchmark tests confirm the target latency and secure handshake standards are met under simulated peak load. Fail: Scan exceeds 3 minutes for 1,000 targets, or an insecure/deprecated TLS handshake is used.	The engine must scale to enterprise-sized domain portfolios without becoming an operational bottleneck, and must not itself introduce a security weakness while probing external endpoints.
NFR-002	The system shall complete at least 99.5% of scheduled daily scan jobs successfully over any rolling 30-day period, automatically retrying transient WHOIS/TLS failures up to 3 times before marking a target as unreachable.	Reliability & Availability	Pass: 30-day scan logs show a completion rate of at least 99.5%, with transient failures retried and resolved automatically. Fail: Repeated scan failures occur without retry or without being logged.	Every alert depends on the underlying scan having run; an unnoticed missed scan silently defeats the entire alerting system, making scan reliability itself safety-critical.

4. UML Use-Case Diagram

The diagram below models both actors, all primary use cases, two <<include>> relationships (the core monitoring use case always performs a WHOIS lookup and a TLS handshake audit), and one <<extend>> relationship (escalation to the Security Officer only occurs conditionally, when an alert is not acknowledged in time or the expiry window is critical).

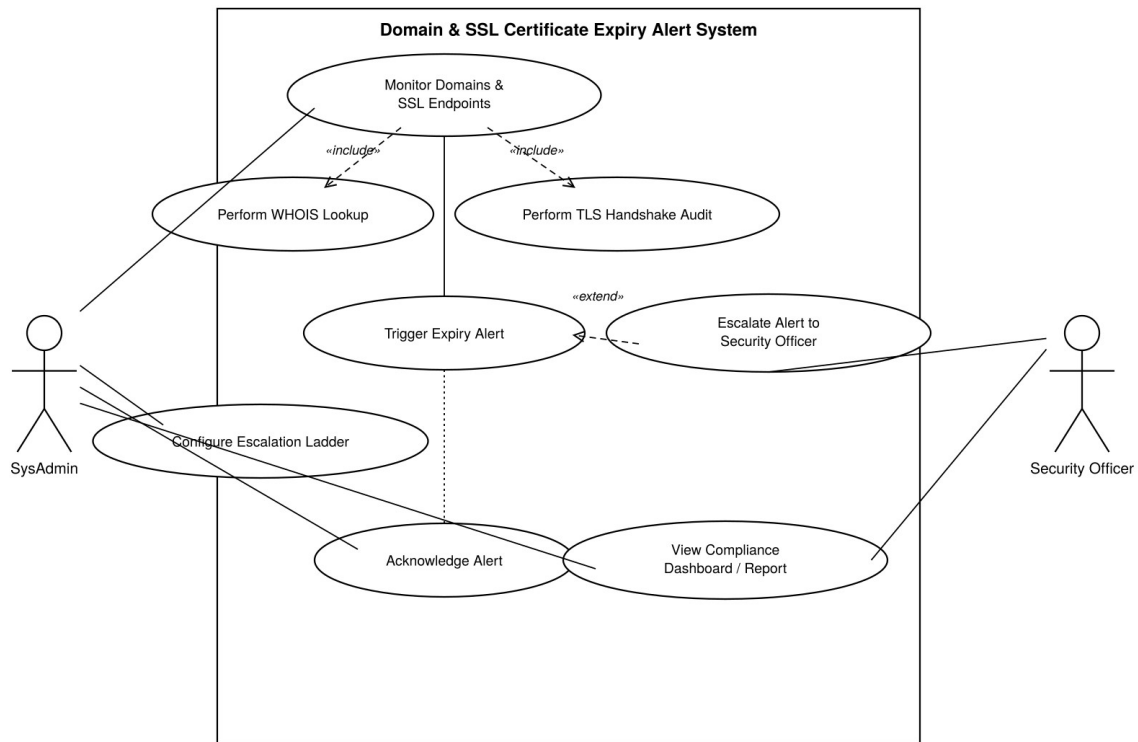


Figure 1: Use-Case Diagram - Domain & SSL Certificate Expiry Alert System

Relationship Summary

- <<include>> - Monitor Domains & SSL Endpoints includes Perform WHOIS Lookup (mandatory sub-step).
- <<include>> - Monitor Domains & SSL Endpoints includes Perform TLS Handshake Audit (mandatory sub-step).
- <<extend>> - Escalate Alert to Security Officer extends Trigger Expiry Alert, activated only under the extension condition (alert unacknowledged past a wait threshold, or days-to-expiry ≤ 3).

5. Use-Case Flow Specification

UC-01: Monitor SSL Certificate & Domain Expiry, and Trigger Expiry Alert

Primary Actor

SysAdmin

Secondary Actor

Security Officer (participates via the <<extend>> alternate flow)

Preconditions

- The list of monitored domains and SSL/TLS endpoints has been configured and the daily scan scheduler is active.
- The system has valid outbound network access to reach each target endpoint's TLS port and the relevant WHOIS registries.
- The escalation ladder and at least one notification channel (e-mail, SMS, or chat-ops webhook) have been configured (FR-003).

Postconditions (on success)

- The certificate-expiry and domain-registration-expiry status of every monitored asset is updated in the system's records.
- For any asset within 30/15/3 days of expiry, an Expiry Alert has been generated, delivered to the appropriate recipients, and logged in the audit trail.

Main Success Scenario

1. The scheduler triggers the daily scan job at the configured time.
2. The system retrieves the current list of monitored domains and SSL/TLS endpoints.
3. [include] Perform TLS Handshake Audit - the system opens a TLS handshake with each endpoint, retrieves its certificate, and extracts the certificate's expiry (NotAfter) date.
4. [include] Perform WHOIS Lookup - the system queries the domain's WHOIS registry and extracts the registration expiry date.
5. The system computes the number of days remaining to expiry for both the certificate and the domain registration.
6. The system compares each value against the alert thresholds (30, 15, and 3 days).
7. If a threshold has been crossed, the system executes Trigger Expiry Alert, creating an alert record with a severity level based on days remaining.
8. The system delivers the alert to the SysAdmin through the channel(s) defined for the current escalation stage.
9. The system records the scan result and any generated alert in the audit log.
10. The use case ends once every monitored asset has been scanned.

Alternate Flow A1 - Escalation to Security Officer (extends step 7/8)

Trigger: The generated alert is still unacknowledged after a configured wait period (e.g., 2 hours), or the computed days-to-expiry is ≤ 3 (Critical severity).

1. A1.1 The system extends the base flow and executes Escalate Alert to Security Officer.
2. A1.2 The alert's severity is upgraded to Critical and the recipient list is widened to include the Security Officer on all configured channels.
3. A1.3 The system records the escalation event, including timestamp and recipients, in the audit log.
4. A1.4 The flow resumes at Main Success Scenario step 9 (log the scan/alert result).